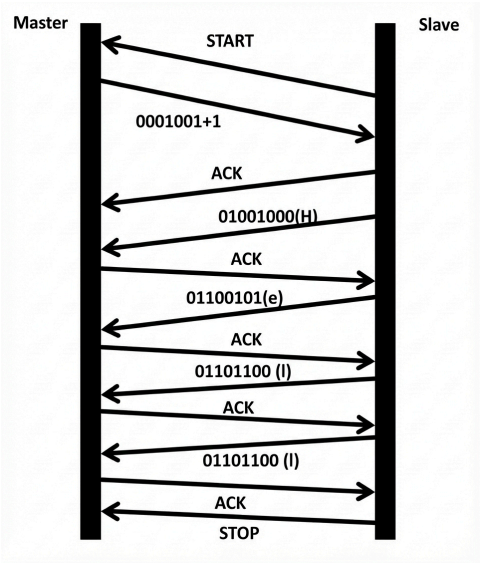
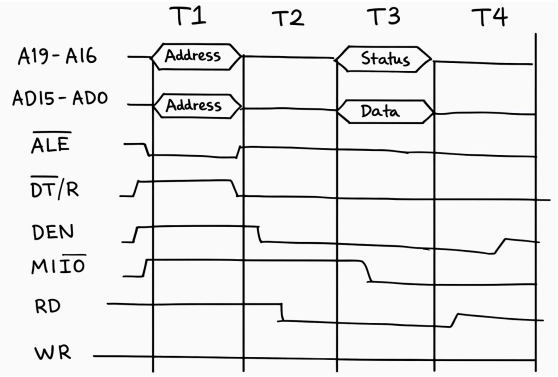
United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam Trimester: Fall 2025
Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
Total Marks: 40 Duration: 2 hours

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (15 Marks)		**[CO5]**
a.	**i)** Tahmid and Sajib are using **two different devices** for **digital-to-analog** conversion. Both of them are obtaining **the same duty cycle**. Tahmid has called **analogWrite(488)** while Sajib has called **analogWrite(122)**. If Tahmid is using a **12-bit DAC** system, **determine which DAC resolution Sajib is using**. [Show all necessary mathematical calculations.]	**4**
	ii) In a microcontroller-based system, an **8-bit ADC and an 8-bit DAC** are used. Both the ADC and the DAC operate with a **reference voltage of 5 V**. An analog input voltage of **3.37 V** is applied to the ADC. The digital output value produced by the ADC is directly used as the argument of the **analogWrite()** function for the DAC system. **Calculate the duty cycle generated by the analogWrite() function.** [Show all necessary mathematical calculations.]	**4**
b.	A student drew an **I²C sequence diagram** (**given below**) where a **master (address: 3H) receives 4 bytes** (‘**HEll**’) from a **slave (address: 9H)**. The teacher pointed out some mistakes in the diagram. **Identify and explain these mistakes.** 	**4**
c.	A company is designing a **low-cost embedded system** for a simple household appliance, such as a digital thermostat. The system only needs to perform basic control operations, like reading temperature sensors and turning a heater on or off. Since the performance is not critical, the designers preferred to reduce hardware complexity and cost. **The system can be built using either Harvard Architecture or Von Neumann Architecture. Which architecture would be more suitable and why?**	**3**
Question 2: Answer all the questions. (10 Marks)		**[CO3]**
a.	Consider the following **initial BIU fetch cycle pattern**: [Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch] Assume **during execution of one instruction**, the BIU fetches **two instructions** into the queue. **Draw** the corresponding BIU and EU cycles for the following cases. **I. 1st instruction** is **‘JUMP <2nd instruction address>’**, **2nd instruction is MOV<address>** and **4th instruction is ‘JUMP <7th instruction address>’**	**3**

	II. 3rd instruction is JUMP <6th instruction address>, 5th instruction is JUMP <8th instruction address> and **6th instruction is a memory read operation.**	**3**
b.	In the Tokyo Jujutsu High lab, the 808x microprocessor is performing an **I/O read** operation. Yuji has drawn the following timing diagram for this scenario. **Identify and redraw only the pins that are incorrect.** 	**4**

Question 3: Answer all the questions. (15 Marks) **[CO4]**

RAM Content

Address	Content		Address	Content
0H	0011 1110		8H	0100 1111
1H	0100 1001		9H	0010 1001
2H	0001 1000		AH	0000 0111
3H	0100 1100		BH	0001 0101
4H	1100 1101		CH	0000 0100
5H	0011 1011		DH	0011 1100
6H	1000 1110		EH	0000 0010
7H	0100 1010		FH	0001 1111

Opcode Table

Mnemonic	Opcode
HLT	1000
OUT	0001
SUB	0011
ADD	0100
LDA	1100

Initially, the value of the **Accumulator Register is 4H, the B Register is 1H,** and the **Output Register is 3H.**

a.	Write **the assembly code** for the instructions stored in **RAM address (0H - 7H)**.	**3**
b.	Determine the **Accumulator and B register** values after executing each instruction in [a].	**3**
c.	**CON =** $E_U \overline{S_U}\, \overline{L_O}\, \overline{L_B}\ \ \overline{L_I}\, \overline{E_I}\, \overline{L_A}\, E_A\ \ E_P\, C_P\, \overline{L_M}\, CE$ **[Observe the pins carefully]** **I.** Write the control words for Execution T-states (T4, T5, T6) to perform the **ADD** instruction using the given controller sequence. **II.** Write the control words for Execution of T-states (T5, T6) to perform the instruction **LDAOUT** (Load the content of memory address to **Accumulator register** and **then** show the value of this register in the **binary display**) using the given controller sequence **with proper explanation.**	**3** **3**
d.	**Initially,** the value of the **Accumulator Register is 4H.** Write the assembly code for the following expression using RAM content: **- 3 - 5 × 4**	**3**